MILITARY SPECIFICATION

DRIFTMETER, ELECTRICALLY DRIVEN GYRO STABILIZED

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers certain design requirements and all performance requirements for one type of optical driftmeter, with an electrically driven gyroscope stabilized reticle, hereafter designated MS28012.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-G-174	Glass, Optical
MIL-C-675	Coating Of Glass Optical Elements (Anti-Reflection)
DOD-D-1000	Drawing, Engineering And Associated List

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-D-3464	Desiccant, Activated, Bags, Packaging Use and Static Dehumidification
MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-P-6131	Plug, Dehydrator, Humidity, Indicating
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-T-7771	Tube, Extension, Gyro Stabilized Drift Meter
MI1-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-O-13830	Optical Component For Fire Control Instrument, General Specification Governing The Manufacture, Assembly And Inspection Of
MIL-L-25142	Luminescent Material, Flourescent
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited.

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810C	Environmental Test Methods
MIL-STD-838	Lubrication Of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1241	Optical Terms And Definitions
MS28012	Driftmeter, Electrically Driven, Gyro Stabilized

DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN5767	Tube-Extension, Gyro Stabilized Drift Meter
AN5768	Rod Assembly-Extension, Drift Meter Prism Control

(Copies of specification, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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American Society For Testing And Material (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials. Materials shall conform to applicable specifications and as specified herein. Materials which are not covered by applicable specifications or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 Metals. Metals shall be of the corrosion-resistant type unless suitably protected to resist corrosion during normal service life.

3.2.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the driftmeter except where magnetic materials are essential.

3.2.3 Dissimilar metals. Where practicable, dissimilar metals, such as defined by MIL-STD-889, shall not be used in intimate contact with each other.

3.2.4 Moisture-and fungus-resistant treatment. Moisture-and fungus-resistant treatment shall be in accordance with the requirements of MIL-E-5400.

3.2.5 Protective treatment. When materials are used in the construction of the driftmeter that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification.

Anodic films and ion vapor deposited aluminum coating shall be in accordance with MIL-S-8625 and MIL-C-83488 respectively. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions, shall be avoided.

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3.2.6 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.2.6.1 Standard parts. Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are out alteration, and provided the corresponding standard part numbers, are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2.7 Recycled and reclaimed materials. Recycled and reclaimed shall be used to the maximum extent possible without jeopardizing the end use of the item.

3.3 Design and construction. The driftmeter shall conform to MS28012. They shall be so constructed that no parts will work loose in service. They shall be built to withstand the strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service, without failure.

3.3.1 Dehydrator plugs. The driftmeter shall be equipment with dehydrator plugs to prevent the fogging of all internal optical surface when the instrument is subjected to extreme temperature changes and high humidities. The dehydrator plugs shall be easily replaceable without disassembly of the instrument and shall be readily accessible for visual inspection of their humidity indication. The dehydrator plugs shall conform to MIL-P-6131 type I or type II unless specified by contracting activity. In addition, each plug shall be provided with a breathing hole 0.030 to 0.040 inch in diameter exposed to the outside air.

3.3.1.1 Each interior airspace which contains or adjoins optical elements shall have direct access to a dehydrator plug through an unobstructed hole. There shall be a negligible flow of air into or out of each such airspace except through the breathing hole provided in each dehydrator plug. The position of the plugs shall not interfere with either the installation or operation of the driftmeter.

3.3.2 Case.

3.3.2.1 Gyro housing cases and tubes. The gyro housing case and tubes shall be made of aluminum alloy, and shall a durable black finish.

3.3.2.2 Reinforcement. The inside of the case and tubes around each connection shall be of sufficient strength to prevent damage to the case or tubes when the connections are tightened during installation of the driftmeter.

3.3.3 Length. The driftmeter shall have provisions for inserting sections of any desired length. Change in lengths up to and inclusive of 90 inches shall require only the insertion of tubing of the required length and rods actuating the trail sighting mechanism.

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Lengths in excess of 100 inches shall require the use of an extension tube made in accordance with MIL-T-7771 in addition to standard sections to make up the required length. In order to insure interchangeability of various standard size tube lengths and extension tubes, the attaching holes in the lower head of the rotating tube and upper end of the outer sleeve of the driftmeter shall be positioned and dimensioned in accordance with AN5767.

3.3.3.1 Prism control rods. Prism control-rod assemblies used in assembling the extension tube to the MS28012 driftmeter shall be in accordance with AN5768-5.

3.3.4 Mounting. The driftmeter shall be designed for mounting vertically in the airplane by means of a mounting flange as shown on MS28012 with additional steadying flanges as specified by the contracting activity, as may be required for support when long extensions are used.

3.3.5 Optical elements. All lenses, mirrors, or reflectors shall be Class I, Grade A, as specified in para 3.3.8.1 of MIL-G-174. Polished surfaces shall conform to the requirements of MIL-O-13830. Dust specks will not be considered sufficient cause for rejection unless they are obvious to the unaided eye when looking directly into the driftmeter eyepiece against a background having the brightness of the sky in average daylight.

3.3.5.1 Coatings. Unless otherwise specified, optics shall be treated with a nonreflecting coating conforming to MIL-C-675.

3.3.6 Optical system. (The following terms are defined in MIL-STD-1241).

3.3.6.1 Aberrations. The optical system shall be free from such aberrations over an angle of 25 degrees as may affect its accuracy when used as a driftmeter.

3.3.6.2 Image. The image shall be erect with respect to centerline of the handle to within 1 degree.

3.3.6.3 Power. The power of the optical system shall be 1 power and 3 power within 10 percent.

3.3.6.4 Exit pupil. Exit pupil shall be not less than 4 millimeters at unit power and maximum length.

3.3.6.5 Curvature of fields. Curvature of fields shall be not more than 1 diopter at low power measured at 10 degrees from the center of the apparent field.

3.3.6.6 Astigmatism. Astigmatism shall be not more than 1-1/2 diopters at low power measured at 10 degrees from the center of the apparent field.

3.3.6.7 Distortion. Distortion shall be not more than 3 percent at low power measured at 10 degrees from the center of the apparent field, with windows and prisms removed.

3.3.6.8 Eyepieces. The driftmeter shall be provided with two eyepieces of 1 power and 3 power which when interchanged shall maintain a suitable seal.

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3.3.6.9 Opening. The opening at the objective end shall be provided with a flat cover glass. Flat is interpreted as a glass surface whose radius of curvature is infinite.

3.3.6.10 True field. The true field at unit power and at all lengths shall be not less than 25 degrees.

3.3.6.11 Warn light. A means shall be provided (warning light) within the field of view to warn the navigator when the instrument is in the caged position. Included on an instruction plate shall be directions to uncage the gyro and wait 2 minutes before using the driftmeter.

3.3.7 Measurement of drift.

3.3.7.1 Mode of operations. The driftmeter shall be designed to measure drift either by sighting vertically down, or by sighting back, to follow signal devices dropped from the aircraft.

3.3.7.2 Angles of operation. The driftmeter shall be capable of measuring drift up to and including 30 degrees right and left and shall be capable of measuring azimuth through 360 degrees.

3.3.7.3 Indication. The indication of drift and azimuth shall agree with the angle of displacement of the drift lines on the reticle to within 1/4 degree at all points of the scale.

3.3.8 Scales.

3.3.8.1 Graduation. The azimuth scale and the line of sight indicating scale shall be graduated in 1-degree divisions.

3.3.8.2 Indicator adjustment. The drift and azimuth scales or the index pointer shall be adjustable for alignment on installation of the meter in the aircraft.

3.3.9 Rotation in azimuth. Rotation in azimuth shall be possible to permit the instrument to be used for measuring bearings or surrounding objects.

3.3.10 Line of sight.

3.3.10.1 Angles required. The line of sight shall be movable from approximately 15 degrees forward to not less than 85 degrees to the rear of the vertical with full field.

3.3.10.2 Scale adjustment. The scale indicating the angle of the line of sight shall be adjustable for exact setting without disassembly of internal operating parts.

3.3.10.3 Detents. Detents shall be provided on the line of sight control at 0.0 degree $\pm$0.1 degree and at 50 degrees and 70.9 degrees $\pm$0.25 degree.

3.3.11 Handle. A handle to control the angle of the line of sight shall be provided on the left side of the instrument. "Left side" refers to observer's left when viewing a 180° azimuth from the side opposite the azimuth indication, with the driftmeter in the installed position."

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3.3.11.1 Gear Ratio. One turn or less of the handle shall move the line of sight through the complete range.

3.3.12 Tangent screw. A tangent screw or other slow-motion device, for rotating the instrument in azimuth, shall be provided. Provision shall be made for rapid rotation of the instrument in azimuth when desired.

3.3.12.1 Scale. One turn of the tangent screw shall rotate the instrument approximately 4 degrees.

3.3.13 Reticle. The apparent thickness and spacing of the reticle lines shall be as shown on Figure 1. The lines used for drift determination shall be alternately heavy and fine. Means shall be provided for use after insertion of a section for exact setting of the reticle lines for collimation and parallelism to the movement of the image when the trail sighting control is operated.

3.3.13.1 Characteristics. The reticle shall be produced on a polished glass disk which is free from flaws, striae, and bubbles which will impair the image quality. The reticle shall be supported on a housing of, and maintained in the vertical by, a self-erecting gyroscope. The reticle shall be so supported that, when the instrument is tilted, the reticle and the image of the ground shall move in the same direction and the same amount.

3.3.13.2 Illumination. The reticle shall be illuminated to give dark-field illumination. This shall be done by a transformation to lamp voltage from the source used to drive the gyroscope. Illumination shall be controlled by means of a rheostat. The range of illumination shall be such as to provide adequate illumination of the grid lines under any lighting condition encountered in normal use.

3.3.14 Gyroscope.

3.3.14.1 Type. The gyroscope shall be of the self-erecting type. It shall be provided with a positive manual caging device and shall have not less than 35 degrees freedom of swing.

3.3.14.2 Stabilization. The gyroscope shall stabilize the reticle to within 1/2 degree of the vertical under conditions simulating normal straight flight.

3.3.14.3 Switch. A switch for starting and stopping the gyroscope shall be provided.

3.3.15 Backlash. The backlash in the gear train that controls drift indications shall be reduced to a minimum by springs or other suitable means.

3.3.16 Filters. A system of polarizing filters shall be provided in the driftmeter to allow a variance of light transmission from maximum to minimum. The positioning of one polarizing filter with respect to the other shall be accomplished by means of knurled knob on the outside of the driftmeter. A clear glass plate shall be incorporated on the same mounting bracket and at 90 degrees to the polarizing filters. A means shall be provided to allow insertion of either the clear glass plate or polarizing filters in the line of sight of the driftmeter.

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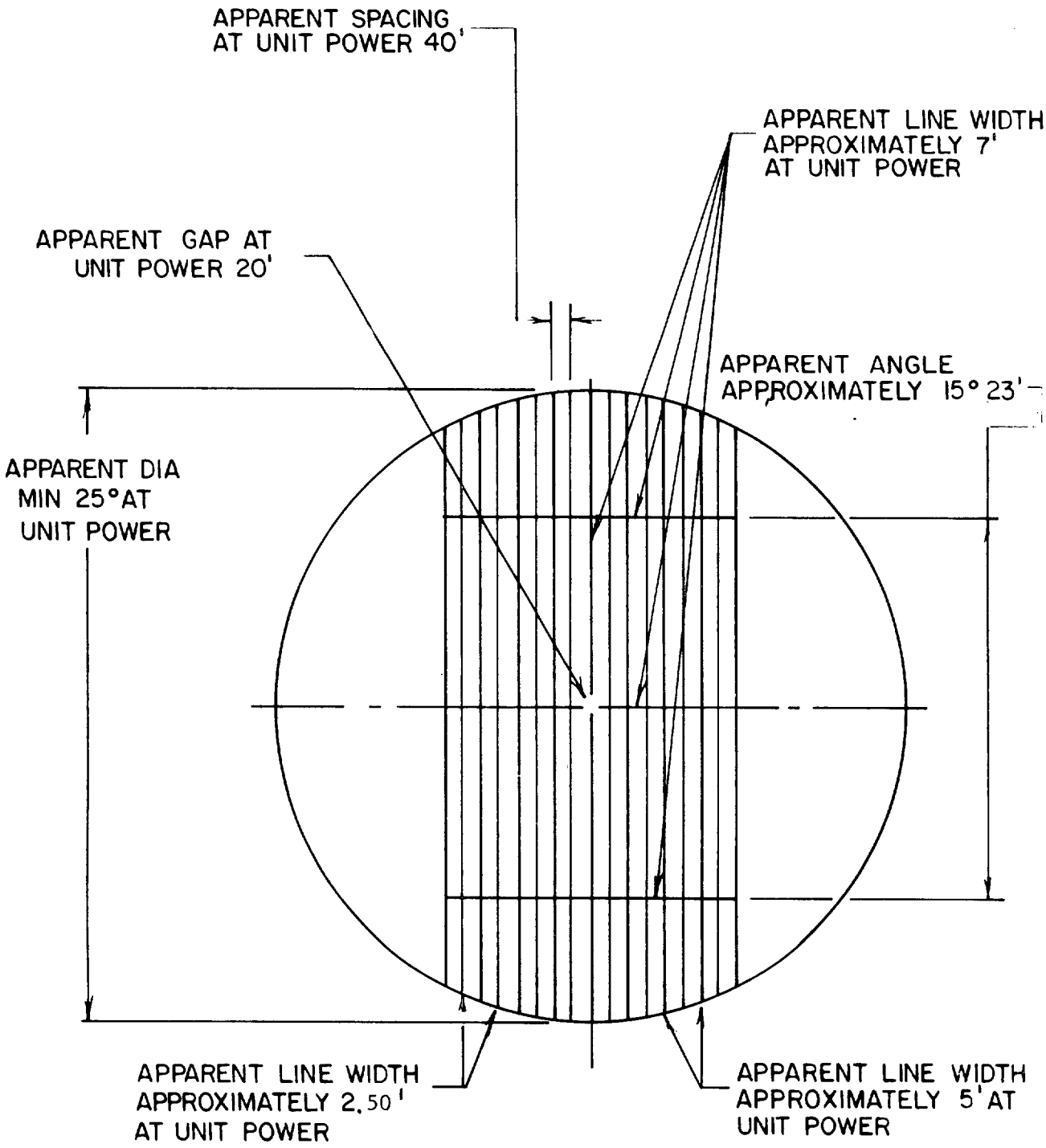


FIGURE 1. Reticle lines

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3.3.17 Electrical receptacles. The electrical receptacles shall be as specified on MS28012. The receptacles shall be so mounted on a stationary part of the driftmeter, that the power leads to the driftmeter shall not rotate when the driftmeter head is rotated. The driftmeter shall operate satisfactorily when power is supplied to either of the two receptacles. Caps shall be furnished to keep the unused receptacle covered at all times.

3.3.18 Screw threads. Screw threads 0.060 or larger in diameter shall be in accordance with MIL-S-7742.

3.3.19 Tube sealing. The entire tube of the 103-inch-length driftmeter shall be sealed against the entrance of water.

3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.5 Weight. The weight of the driftmeter with a tube of minimum length, shall not exceed 16 pounds.

3.6. Finish.

3.6.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625.

3.6.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.7 Lubrication. Lubrication shall be accomplished in accordance with MIL-STD-838, except that the selection of lubricants used shall be approved by the contracting activity.

3.8 Operational performance. The driftmeter shall satisfy the performance requirements as specified in section 4 when subjected to the applicable tests.

3.8.1 Power requirements. The driftmeter shall be capable of operation from the following sources and shall not require power in excess of the amount indicated:

A-c sources:

115/120 volts, 380-420 HZ, 3-phase, 100VA, 3-wire, 115-volt line-to-line system.

115/200 volts, 380-420 HZ, 3-phase 100VA, 4-wire, 115-volt line-to-neutral system.

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3.8.1.1 Voltage limits. The driftmeter shall be designed to meet the performance requirements specified herein for a source voltage of 110 to 120 volts. In addition, the equipment shall operate at voltages within the limits of 102 to 110 and 120 to 124, but this operation may be be reduced performance and shall not damage the equipment. If practicable, (i.e., no excess weight or form factor penalty), the equipment shall be designed to meet the performance requirements over the range of 102 to 124 volts

3.8.1.2 Power supply frequency deviation. The equipment shall be of such design that it will not be damaged by power supply frequencies below the minimum specified and shall automatically resume normal operation when the frequency returns within limits; however, there are no requirements on equipment performance at frequencies below the lower established operating limit. Down to 200 HZ the voltage shall be considered as maintained within the operating limits. Below 200 HZ the voltage shall be considered as varying proportionally to the frequency, with zero voltage existing at zero frequency.

3.9 Special tools. The design of the driftmeter shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.10 Markings. All markings shall be sufficiently durable to withstand usage encountered in service.

3.10.1 Numerals. Numerals shall distinctly indicate the graduation to which each applies. Any confusion resulting in doubt as to which graduation the numeral applies will be cause for rejection. When several numerals are used in one group, the space between the numerals shall be approximately 0.016 inch.

3.10.2 Scales. The driftmeter scales shall be finished in luminescent fluorescent material in accordance with MIL-L-25142. The dimensions of the markings shall be in accordance with Table I.

TABLE I. Scale markings dimensions.

Markings	Height or length .016 (inch)	Width of line or graduation ±0.005 (inch)
1-Degree graduations	0.094	0.016
1-Degree graduations	0.094	0.016
5-Degree graduations	0.125	0.016
10-Degree graduations	0.188	0.016
Lettering "Drift"	0.094	0.016
Lettering "R" and "L"	0.125	0.016

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3.10.3 Operating instructions. The following instructions shall be legibly and permanently marked on a readily visible plate which shall be securely attached to the upper housing of the driftmeter.

3.10.4 Instructions for use. The instructions for operating the driftmeter relating to starting, stopping, and precautions in use shall be marked on the plate.

3.10.5 K factor. The K factor for each driftmeter is to be determined by the manufacturer in accordance with the procedure as approved by the contracting activity. The instructions for use of the K factor for determining ground speed by zero trail angle method shall be marked on the plate.

3.10.6 Caution marking. The following information shall be marked on the plate:

CAUTION
WHEN POWER IS OFF KEEP
GYRO CAGED

3.11 Identification of product.

3.11.1 Nameplate. The nameplate shall be securely attached to the exterior of the case and shall be legibly and durably marked with the information listed below. The information marked in the spaces on the nameplate shall be in accordance with MIL-STD-130.

DRIFTMETER, ELECTRICALLY DRIVEN GYRO STABILIZED
MS28012 (Proper dash No.)
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
US

3.12 Installation instructions. Unless otherwise specified, the contractor shall furnish with each instrument, one printed copy of instructions, with illustrations and diagrams if necessary, covering the installation of the instrument. Prior to printing, two copies shall be furnished to the contracting activity for approval. The instructions shall be printed on 8-1/2 by 11 or 17 by 11-inch paper.

3.12.1 Envelope. When applicable, an envelope furnished by the contractor containing the installation instructions shall be packed with each instrument, and shall be marked with the following information:

IMPORTANT
THIS ENVELOPE CONTAINS INSTALLATION
INSTRUCTIONS

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3.13 Spare parts.

3.13.1 Dehydrator plug spares. Each driftmeter shall be supplied with 8 spare dehydrator plugs conforming to the requirements of paragraph titles Dehydrator plugs (3.3.1) with all holes sealed and enclosed in a moisture-proof container.

3.13.2 Lamp. Each driftmeter shall be supplied with one spare lamp. Provisions shall be made for carrying the spare lamp on the body of the driftmeter in an easily accessible location.

3.14 Workmanship. All details of workmanship shall be in accordance with MIL-STD-454 requirement 9, high-grade optical instrument manufacturing practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of driftmeters shall be classified as follows:

a. First article inspection (see 4.3)

b. Acceptance tests (see 4.4)

4.3 First article test.

4.3.1 Sampling instructions. The first article test samples shall consist of two driftmeters of each manufacturer's part number. Samples shall be tested at the places designated in the purchase document; see 6.2.d.

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4.3.2 Tests. The first article tests of the driftmeter shall consist of all the tests of this specification, as described under Test Methods. Pre-production tests may, at the option of the contracting activity, be supplemented with tests under actual service conditions. The tests shall be conducted in the order listed in paragraph 4.6.

4.4 Acceptance tests. Acceptance tests shall consist of Individual tests and Sampling tests.

4.4.1 Individual tests. Each driftmeter submitted for acceptance under contract shall be subjected to the following tests, as described under Test Methods.

a. Examination of product.
b. Moving parts.
c. Line of sight.
d. Rotation.
e. Reticle.
f. Starting torque.
g. Erection system.
h. True field.
i. Focus.
j. Sealing.
k. Waterproof.

4.4.2 Sampling tests. A minimum of 3 driftmeters, or one from every 100 driftmeters or fraction thereof, submitted for acceptance on a contract (whichever is greater) shall be subjected to the following tests, as described under Test Methods:

a. Low temperature operation.
b. High temperature operation.
c. Preservation (See 4.7).

4.4.3 Rejection and retest. When tests are specified on a quantity of driftmeters that are selected as representative of a certain lot, and one or more of this number fails to meet the requirements, additional driftmeters of the lot represented shall be tested immediately to determine the cause of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the driftmeter. When the cause of failure has been determined, the contractor shall incorporate the changes necessary to correct the defects in the remaining driftmeters on the order.

4.4.3.1 Rejected driftmeters. Rejected driftmeters either shall be replaced with other driftmeters or shall be reworked to correct the defects, after which, all specified tests shall be repeated. Rejected driftmeters shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

4.5 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5.1 of this specification.

4.5.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of approximately 25°C (77°F).

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When tests are made at atmospheric pressure and temperature substantially different from the above values, proper allowance shall be made for the difference from specified conditions.

4.6 Test methods

4.6.1 Examination of product. Each driftmeter shall be carefully examined to determine conformance with the requirements of this specification not covered by tests.

4.6.2 Moving parts. All moving parts shall be movable within their operational limits. They shall operate freely and without excessive backlash.

4.6.3 Line of sight. The indication of the angle of the line of sight with the gyro caged shall be noted. It shall agree with the angle within 0.50 degree at all points of the scale.

4.6.4 Rotation. The meter shall be rotated by means of the tangent screw. It shall turn freely and smoothly in the mounting bearing and shall be readily rotatable by means of the tangent screw. One turn of the tangent screw or operating knob shall rotate the sight approximately 4 degrees.

4.6.5 Reticle. The trail sighting control shall be operated and the movement of the image noted. When the control is operated, the image shall move parallel to the drift lines to within 0.25 degree throughout the range.

4.6.6 Starting torque. The starting torque of the gyroscope motor shall be sufficient to reach normal operating speed from zero revolutions per minute within 5 minutes after starting when in the caged.

4.6.7 Erection system. The erection system shall be checked. The erection system shall produce a recovery rate of not less than 2 degrees per minute nor more than 5 degrees per minute at 10 degrees from the vertical. The gyro shall come to vertical within 15 minutes from a 10-degree tilt position.

4.6.8 True field. The true field at unit power shall be noted. It shall be not less than 25 degrees.

4.6.9 Focus. The image shall be focused at high power. The reticle lines shall be in sharp focus.

4.6.10 Sealing. With the dehydrator plug holes sealed, a vacuum of 20 inches of mercury shall be applied to the sealed driftmeter. The vacuum shall not drop off more than 10 inches of mercury in 1 minute when the supply is shut off.

4.6.11 Waterproof test. Each driftmeter having a 103-inch tube length shall be tested for watertightness by immersing the complete tube length vertically in fresh water for a period of 1 hour with no visible evidence of leakage into the tube.

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4.6.12 Low temperature operation. The driftmeter shall be subjected to a temperature of $-55^{\circ} \pm 2^{\circ}C$ ($-67^{\circ} \pm 3.6^{\circ}F$) for a period of 4 hours, and then tested for freedom of motion of all moving parts. No tests are to be made for the accuracy of the instrument. It shall not take the gyro rotor more than 25 minutes to attain a serviceable speed at this temperature. Minor changes in the appearance of the cement in cemented optical elements produced during the low temperature tests shall not be considered cause for rejection.

4.6.13 High temperature operation. The driftmeters shall be subjected to a temperature of $70^{\circ} \pm 2^{\circ}C$ ($158^{\circ} \pm 3.6F$) for a period of 2 hours. The driftmeter shall operate without any indication of sticking, erratic action, or fogging while at this temperature.

4.6.14 Fogging. Prior to the start of this test, the dehydrator plugs shall be installed for a period of 10 hours, and means shall be provided to insure having dry air in the unit before the start of the test. The driftmeter with dehydrator plugs properly installed shall be placed in an atmosphere of 95 to 100 percent relative humidity at $22^{\circ} \pm 2^{\circ}C$ ($71.6^{\circ} \pm 3.6^{\circ}F$) for a period of 1 hour. The instrument shall then be immediately subjected to a temperature of $40^{\circ} \pm 2^{\circ}C$ ($-40^{\circ} \pm 3.6^{\circ}F$) and a pressure equivalent to 30,000 feet altitude for 1.2 hour. The driftmeter shall then be subjected to standard room temperature and pressure and so placed between heating elements that the lower telescope case receives the direct heat radiation. As the temperature of the driftmeter rises, there shall be no evidence of fogging of the internal optical elements.

4.6.15 Low temperature exposure. The driftmeters shall be subjected to a temperature of $-65^{\circ} \pm 2^{\circ}C$ ($-85^{\circ} \pm 3.6^{\circ}F$) for a period of 48 hours. After this period, the driftmeter shall be raised to a temperature of $-55^{\circ} \pm 2^{\circ}C$ ($-67^{\circ} \pm 3.6^{\circ}F$) for a period of 8 hours and then tested for freedom of motion of all moving parts. No tests are to be made for the accuracy of the instrument. It shall not take the gyro rotor more than 25 minutes to attain a serviceable speed at this temperature. Minor changes in the appearance of the cement in cemented optical elements produced during the low temperature tests shall not be considered cause for rejection. This test may be combined with the Low temperature operation test at the discretion of the testing activity.

4.6.16 Salt fog. The driftmeter shall be subjected to the salt fog test per Method 509.1, Procedure I of MIL-STD-810C. At the end of the salt fog test, the driftmeter shall be subjected to, and shall pass, the individual tests specified herein.

4.6.17 Fungus. The driftmeter shall be subjected to the fungus test per Method 508.1, procedure I of MIL-STD-810C. The test shall be conducted at the conclusion of first article tests. At the end of the test period, the driftmeter, the driftmeter shall be carefully examined to ascertain that no fungus growth has occurred. There shall be no damage that adversely affects operation of the driftmeter.

4.6.18 Vibration. The driftmeter shall be subjected to the vibration test of MIL-STD-810C, Method 514.2, Procedure I, Table 514.2-II, for equipment category b.1 (except delete reference to figure 514.2-2), with the driftmeter mounted in its normal operating position.

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The vibration spectrum shall consist of a constant, sinusoidal vibration of 0.4 inch double amplitude from 5 Hz to 10 Hz, then a constant 2g from 10 Hz to 500 Hz. After the vibration test, the driftmeter shall meet the requirements set forth under individual tests. In addition, no looseness in the mechanism or damage to any part of the driftmeter shall result from the vibration.

4.6.19 Humidity. The driftmeter shall be subjected to the humidity test per Method 507.1, Procedure I of MIL-STD-810C.

4.6.20 High temperature exposure. The driftmeter shall be subjected to a temperature of 71° ±2°C (160° ±3F) for a period of 24 hours. At the end of the 24-hour period and while still at the high temperature, the driftmeter shall operate without indication of sticking or erratic action. After the chamber has been returned to room temperature and the equipment has saturated to room temperature for a minimum period of 4 hours, the driftmeter shall meet the Individual test requirements specified herein.

4.7 Preservation. Tests of methods of preservation shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. Packaging

5.1 Preservation. Preservation shall be level A, C, or Industrial as specified. (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Meter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Meter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each meter shall be packaged in quantity unit packs in one each in accordance with Method IC-1 of MIL-P-116. Each meter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G-s established for the meter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each meter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of the meter shall be in accordance with ASTM D3951.

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5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Meter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2 Level B. Meter packaged as specified in 5.1.1.1 shall be packed in shipping containers conforming to PPP-B-636, Class weather-resistant, unless specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords, adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged meter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2) interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The gyro stabilized driftmeter is intended for use by the navigator on aircraft for measuring the angle between the heading of the aircraft and its actual path over the surface of the earth and for obtaining relative bearings on surrounding objects.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. MS Part No. of the meter desired, whether additional steadying flanges are necessary, the type of nonreflecting coating on the optics, type of dehydrator plus and whether the accessories are to be operated from 24-volt direct current.
c. The type of packaging and packing required.
d. Where the first article test samples shall be sent and instructions concerning the submittal of the test reports.

6.3 First article. When a first article is required, the item shall be inspected and tested. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

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6.4 Subject term (key word) listing.

Driftmeter
Electronic
Gyro
Optical

Custodians:
AIR FORCE -99
NAVY -AS

Preparing Activity:
AIR FORCE -71

Project No:
6605-0366

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-7106C	Driftmeter, Electrically Driven GYRO Stabilized

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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